MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, SERVO AMPLIFIER TEST INPUT

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-60898 comprise the complete requirements for the Servo Amplifier Test Input Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-60898	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10554194	Electronic Component Assembly, Servo Amplifier Test Input

PACKAGING DATA SHEET

10554194	Packaging of Circuit Card Assembly, Servo Amplifier Test Input

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4931

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3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F10554194 and drawings pertaining thereto.

3.2' General specification. The assembly shall meet the requirements of MIL-C-60898.

3.3 Environmental service conditions. See MIL-C-60898.

3.3.1 Performance at temperature extremes. For the conditions of 3.3.2 of MIL-C-60898, the requirements of 3.4 shall be met with exceptions as specified in the following subparagraphs.

3.3.1.1 Diodes, reverse current. As specified in 3.4.1,4 except the current shall be no greater than 0.1 ma between the pins specified.

3.3.1.2 Diodes, forward drop. As specified in 3.4.1.5 except the voltage drop shall be 1.0 ± 0.5 vdc between the pins specified.

3.3.1.3 Capacitance. As specified in 3.4.2 except the capacitance between pin 27 and pin 1 shall be 0.1 μf ± 15 percent.

3.3.1.4 Resistance. As specified in 3.4.3 except the resistance shall be ± 3 percent and ± 10 percent for the values specified.

3.4 Electrical performance

3.4.1 Diode network

3.4.1.1 Pin 6 output. With the input circuitry connected to pin 6 as shown in Figure 1, the voltage at pin 6 shall be less than +3.55 vdc but greater than +3.00 vdc.

3.4.1.2 Pin 12 output. With the input circuitry connected to pin 12, as shown in Figure 1, the out put voltage at pin 12 shall be less than +3.55 vdc but greater than +3.00 vdc.

3.4.1.3 Pin 9 output. With the input circuitry connected to pin 6 or pin 12, as shown in Figure 2, the output voltage at pin 9 shall be less than +5.85 vdc but greater than +5.40 vdc. With the input circuitry connected to pin 6 or pin 12, as shown in Figure 2 but with reversed polarity, the output voltage at pin 9 shall be - 10 ± 0.1 vdc.

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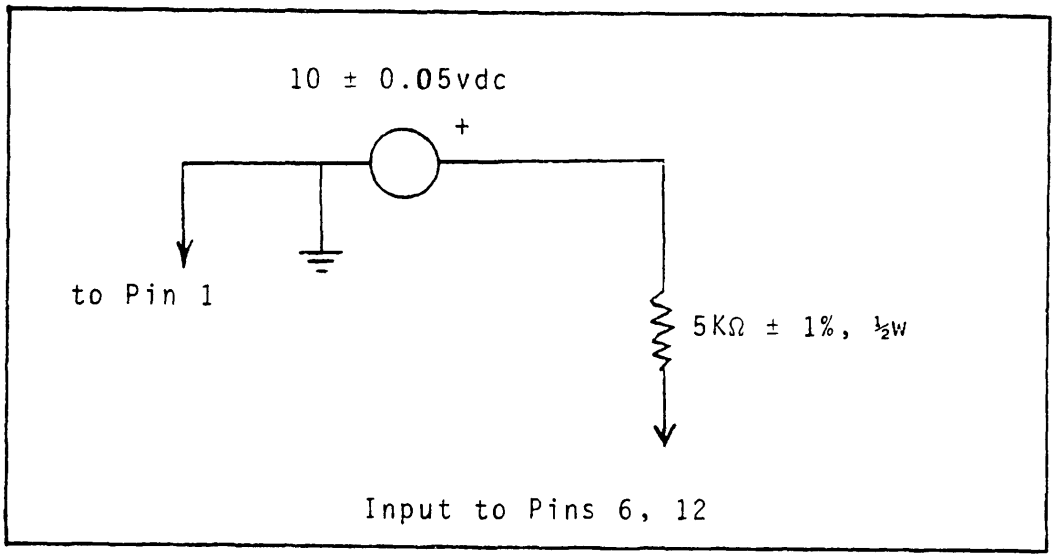


Figure 1

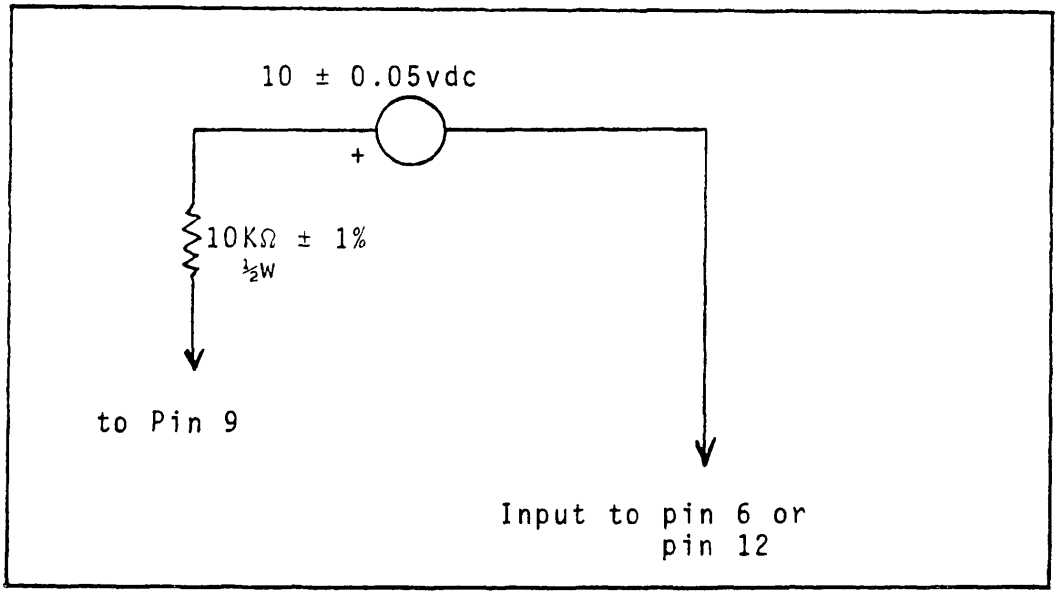


Figure 2

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3.4.1.4 Diodes, reverse current. With +28.0 ± 1.0 vdc applied between each of the two pins specified in Table I, there shall be +0.02 milliamperes or less current between the two pins.

Table I

Reverse Current

Pin (+) (28 vdc)	Pin (-) (GND)
23	1
26	1

3.4.1.5 Diodes, forward drop. With a current of 1.0 ± 0.1 amperes applied between each of the two pins, specified in Table II, the voltage between these pins shall not be greater than 0.9 ± 0.3 vdc.

Table II

Pin (+)	Pin (-)
1	23
1	26

3.4.2 Capacitance. The capacitance between pin 27 and pin 1 shall be 0.1 uf ± 10 percent.

3.4.3 Resistance. The resistance between pins shall be as specified in Table III.

Table III
Terminal Resistance

Pins	Resistance
3, 4	3,320 ohms (± 1 percent)
4, 6	302,210 ohms (± 1 percent)
15, 16	3,320 ohms (± 1 percent)
12, 15	302,210 ohms (± 1 percent)
21, 22	1,000 ohms (± 5 percent)
22, 1	150 ohms (± 5 percent)
17, 18	5,490 ohms (± 1 percent)
19, 20	5,490 ohms (± 1 percent)

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3.4.4 Pulse transformers. With the terminations to the board connected as shown in Figure 3, the output between pin 11 (+) and pin 8 (-) shall be as shown in Figure 4 when switch A of Figure 4 is placed in position 2.

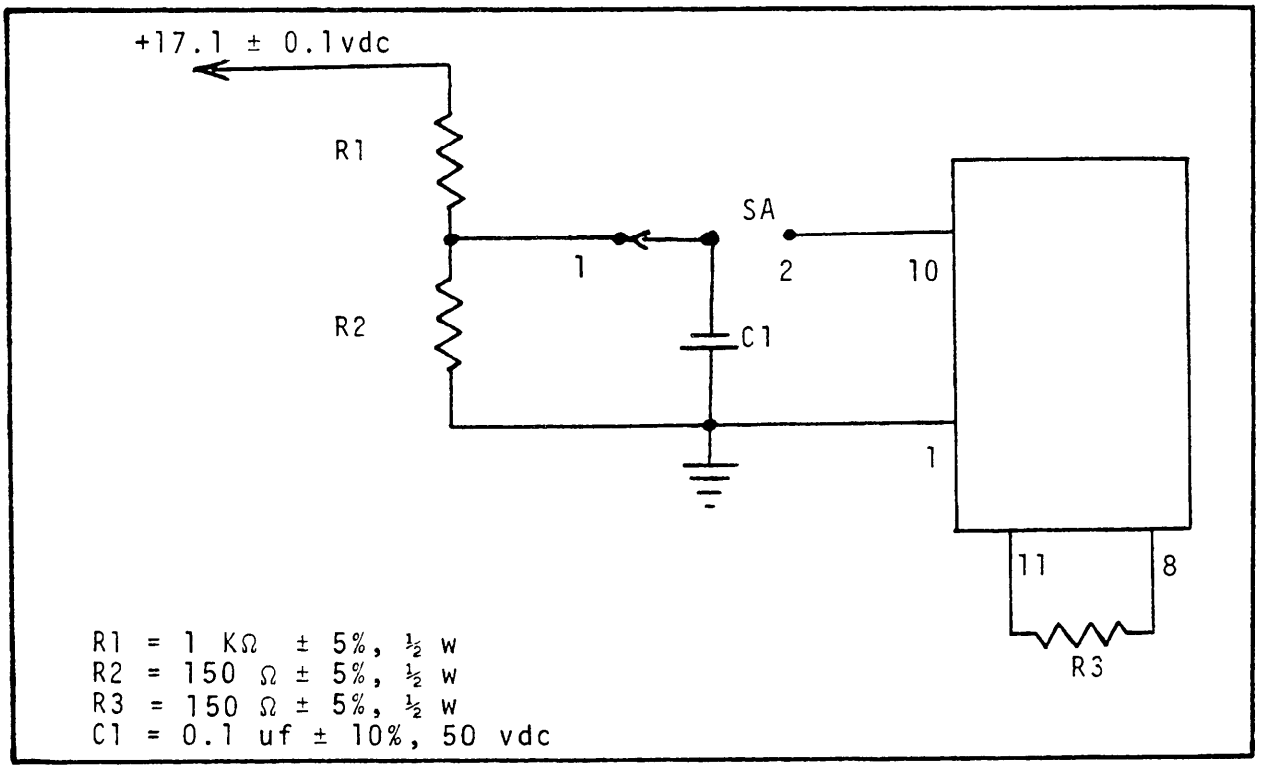


Figure 3

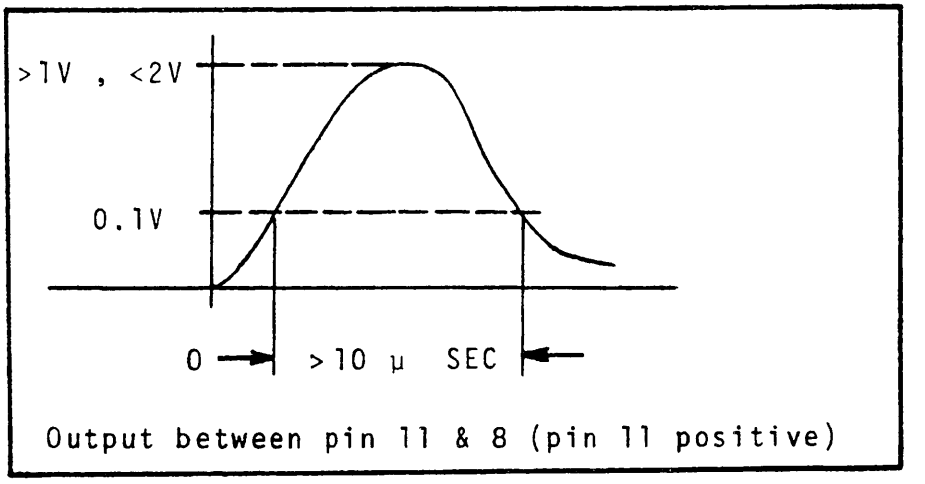


Figure 4

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4. QUALITY ASSURANCE PROVISIONS

4.1 General, The quality assurance provisions of MIL-C-60898 shall apply to the assembly covered by this detail specification,

4.2' First article inspection. As specified in MIL-C-60898.

4.3 Control test sample. As specified in MIL-C-60898.

4.4 Functional tests. The following characteristics shall be 100 percent tested in accordance with MIL-C-60898.

Characteristic	Requirement	Procedure
Diode networks	3.4.1	4.5.2.1
101 Pin 6 output	3.4.1.1	4.5.2.1.1
102 Pin 12 output	3.4.1.2	4.5.2 .1.2
103 Pin 9 output	3.4.1.3	4.5.2.1.3
104 Diodes, reverse current	3.4.1.4	4.5.2.1.4
105 Diodes, forward drop	3.4.1.5	4.5.2.1.5
106 Capacitance	3.4.2	4.5.2.2
107 Resistance	3.4.3	4.5.2.3
108 Pulse transformers	3.4.4	4.5.2.4

4.4.1 Inspection equipment. As specified in MIL-C-60898 and as below in 4.4.1.1.

4.4.1.1 Test equipment. In conjunction with 4.4.1, the following equipment shall be utilized in the performance of the applicable tests as specified in 4.5.

Item	Description
1. POWER SUPPLY	+10.00 ± 0.05 vdc, no less than 500 ma, no greater than 50 mv rms ripple.
2. POWER SUPPLY	+17.1 ± 0.1 vdc, no less than 500 ma, no greater than 100 mv rms ripple.
3. OSCILLOSCOPE	Voltage, time accuracy, and resolution no less than 5 percent. Rise time no greater than 100 nanoseconds. Sweep periods from one microsecond per centimeter to one second per centimeter. Amplitude scales from 0,01 to 10.0 volts per centimeter. Automatic triggering,

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4.	DIGITAL VOLTMETER	Accuracy and resolution no less than 0.2 percent from 0.0090 to 99.0 vdc. Input impedance no less than 10 megohms on all voltage scales,
5.	AMMETER	0.0 to 5.0 ampere with 2 percent resolution and accuracy at 0.02 ma and 1.0 ampere.
6.	ASSORTED COMPONENTS	
	RESISTOR	5 kilohm ± 1 percent, 0.5 watt
	RESISTOR	1 kilohm ± 5 percent, 0.5 watt
	RESISTOR	10 kilohm ± 1 percent, 0.5 watt
	RESISTOR (2 ea)	150 ohm ± 5 percent, 0.5 watt
	CAPACITOR	0.1 uf ± 5 percent, 50 vdc
	SWITCH	5.0 ampere, single pole-double throw
7.	POWER SUPPLY	Variable 0.0 to 30 vdc, 0.0 to 1.5 amperes.
8.	CAPACITANCE BRIDGE	Capacitance display from 0.005 uf to 100 uf, no less than 1.0 percent accuracy and resolution in ac/dc operation.

4.5 Test Methods and Procedures

4.5.1 Environmental. As specified in MIL-C-60898 except the requirements of 3.3.1 herein shall be verified during temperature extremes.

4.5.2 Electrical tests

4.5.2.1 Diode networks

4.5,2. 1.1 Pin 6 output. Connect the ASSORTED COMPONENTS, item 6, and POWER SUPPLY, item 1 of 4.4.1.1, to the card as shown in Figure 1. Measure the dc voltage between pin 6 (+) and pin 1 (-) with the DIGITAL VOLTMETER, item 4 of 4.4.1.1, to verify conformance with the requirements of 3.4.1.1.

4.5.2.1.2 Pin 12 output. Connect the ASSORTED COMPONENTS, item 6, and POWER SUPPLY, item 1 of 4.4.1.1, to the card as shown in Figure 1. Measure the dc voltage between pin 12 (+) and pin 1 (-) with the DIGITAL VOLTMETER, item 4 of 4.4.1.1, to verify conformance with the requirements of 3.4.1.1.

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4.5.2.1.3 Pin 9 output. Connect POWER SUPPLY, item 1 of 4.4 1.1 in series with the 5 kilohm resistor of ASSORTED COMPONENTS, item 6 of 4.4.1.1 as shown in Figure 1. Apply the series combination between pin 6 and pin 9 and then between pin 12 and pin 1. Repeat for both polarities. Using the DVM, item 4 of 4.4.1.1 measure the voltage between pin 9 and pin 6 or pin 12 for each of the four connections to verify the requirements of 3.4.1.3.

4.5.2.1.4 Diodes, reverse current. Connect the AMMETER, item 5, and POWER SUPPLY, item 7 of 4.4.l.l, as shown in Figure 5. Adjust the POWER SUPPLY for an output of 28.0 ± 0.1 vdc and verify that the AMMETER indicates 0.02 ma or less for each connection specified in 3.4.1.4.

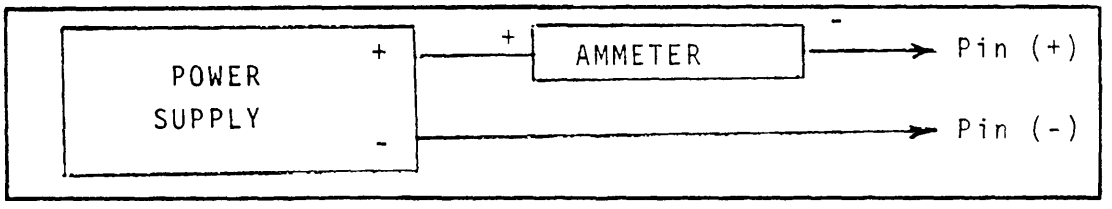


Figure 5

4.5.2.1.5 Diodes, forward drop. Connect the AMMETER, item 5, and POWER SUPPLY, item 7 of 4.4.1.1, as shown in Figure 5. Adjust the POWER SUPPLY such that the AMMETER indicates 1.0 ± 0.1 ampere. Connect the AMMETER and POWER SUPPLY to the card as specified in 3.4.3. Verify, with the DIGITAL VOLTMETER, item 6 of 4.4.1.1, that the voltage is 0.9 ± 0.3 vdc.

4.5.2.2 Capacitor. Using the CAPACITANCE BRIDGE, item 8 of 4.4.1.1, measure the capacitance between pin 27 and pin 1 to verify conformance with 3.4.2.

4.5.2.3 Resistance. Measure the resistance between the pins specified in Table II with the DIGITAL VOLTMETER, item 4 of 4.4.1.1, to verify conformance with the requirements of 3.4.3.

4.5.2.4 Pulse transformer. Connect the ASSORTED COMPONENTS, item 6, and POWER SUPPLY, item 2 of 4.4.1.1, as shown in Figure 3. Connect the OSCILLOSCOPE, item 3 of 4.4.1.1, to pin 11 (+) and pin 8 (-). Adjust the OSCILLOSCOPE to trigger on an internal positive going signal. Observe the trace on the OSCILLOSCOPE and place switch A in position 2. Verify that the waveform on the OSCILLOSCOPE conforms to 3.4.4 and Figure 4.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with applicable requirements of MIL-C-60898.

6. NOTES

6.1 The notes specified in MIL-C-60898 are applicable to this specification.

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Preparing Activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES , NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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